

From sample to reads: designing an experiment

How much sequencing is enough for the stated question?

By the end, you should be able to...



Explain reads, read pairs, quality and depth.



Calculate nominal and usable average depth with units.



Choose a feasible design without confusing more reads with more replicates.

Reconnect with the biology

DNA sequencing measures sequence fragments; RNA sequencing samples molecules derived from RNA.

Think silently -> explain to a neighbour -> check the measurement.

A library is a prepared collection of molecules

01

Samples are processed into a sequencing library.



02

A read is a reported sequence from that library.



03

PCR and technical duplication can create repeated observations of the same original material.

Short and long reads are design choices

What must be resolved?

What can be measured?

Longer reads can span larger structures or transcript isoforms.

Short-read workflows can be useful for many counting and variant questions.

Choose using the assay question, sample quality and required evidence, not a vendor ranking.

FASTQ links bases to quality

```
@toy_read  
ACGT  
+  
IIII
```

A FASTQ record contains an identifier, sequence, separator and quality string.

The number of base qualities should match the sequence length.

Quality is about base-call error, not whether the biological conclusion is true.

Phred quality is a log-scale error measure

Q	Error probability
20	1 / 100
30	1 / 1,000
40	1 / 10,000

$Q = -10 \log_{10}(p_{\text{error}})$.

Q20 corresponds to $p_{\text{error}} = 0.01$; Q30 to 0.001.

A quality score describes a model of base-call error; quality calibration matters.

Average depth is a budget calculation

$$C = \text{sequenced bases} / \text{target bases}$$

State the units and the assumptions.

For N single reads of length L over target size G: $C = N \times L / G$.

For read pairs, count both sequenced ends in the base budget.

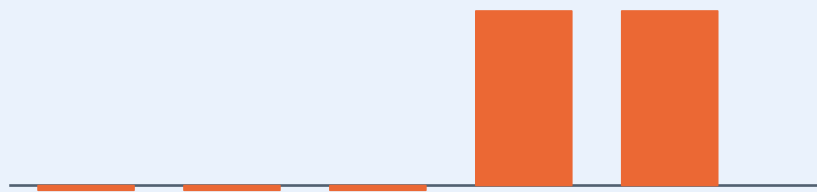
This is nominal average depth; overlapping reads and duplicates complicate unique evidence.

Average is not the same as everywhere

Synthetic: both means = 20x



Uniform coverage



Uneven coverage

The same average depth can hide very different coverage distributions.

Report the fraction of target positions reaching a threshold.

No universal depth cutoff guarantees a clinically usable result.

Usable yield is smaller than purchased yield

**usable depth = nominal
depth x usable fraction**

State the units and the assumptions.

A toy usable fraction combines retention and duplicate assumptions.

Keep assumptions explicit; duplicate rate can depend on library complexity.

Sensitivity analysis: compare what happens when usable yield is lower than expected.

Design Notebook 1: the decision, not a shopping list

01

Choose target,
independent samples
and a sequencing
budget.

>

02

Compare two designs,
including a control and
one failure scenario.

>

03

Defend a limited
conclusion; all quoted
classroom costs are
hypothetical.

Worked example: one million read pairs

1,000,000 pairs, 150 bases per end, target size 5,000,000 bases, usable fraction 0.80.

Predict the result and one limitation before calculating.

Work through the reasoning

Sequenced bases = $1,000,000 \times 2 \times 150 = 300,000,000$.

Nominal average depth = $300,000,000 / 5,000,000 = 60x$.

Toy usable depth = $60 \times 0.80 = 48x$; this is not guaranteed depth at every base.

Pause: make a defensible claim

Two libraries from the same donor each have 60x depth. How many donors were measured? What does sequencing both libraries more deeply fail to fix?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

One donor. More technical observations do not add independent donors or establish between-donor variation.

Today's practical

W05 / Project D1: select one of three fully specified budget scenarios, calculate depth, inspect a simulated coverage distribution and write a design justification.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 library
- 2 read pair
- 3 quality
- 4 depth
- 5 replication

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: derive the ideal Poisson zero-coverage probability $\exp(-C)$, then explain why real coverage may violate that model.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S03 SAM/BAM, VCF and BED specifications, maintained by the HTS community.

S10 DESeq2: Analyzing RNA-seq data with DESeq2, release vignette.